



Lecture 3:

How do we build electronics?

ECE 340 – Electronic Circuits I Sec. 1

Shuva Mitra

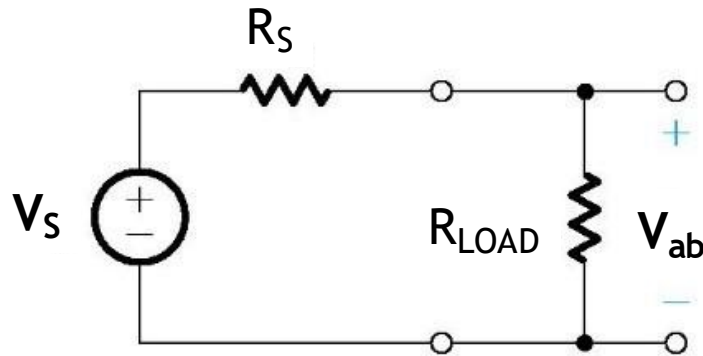
9/10/2025

(Textbook: Finish Chapter 1,
Start Chapter 3)



Department of Electrical
and Computer Engineering
UNIVERSITY OF WISCONSIN-MADISON

How to make “ideal” sources



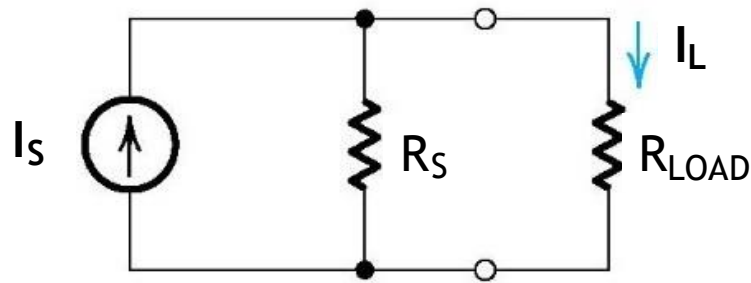
What do we need to do to make $V_{ab} = V_S$?

$$V_{ab} = \frac{R_{LOAD}}{R_{LOAD} + R_S} * V_S$$

$$1 = \frac{R_{LOAD}}{R_{LOAD} + R_S}$$

true only if $R_{LOAD} = R_{LOAD} + R_S$
so $R_S \ll R_{LOAD}$ Ideally, $R_S \rightarrow 0$

How to make “ideal” sources



What do we need to do to make $I_L = I_s$?

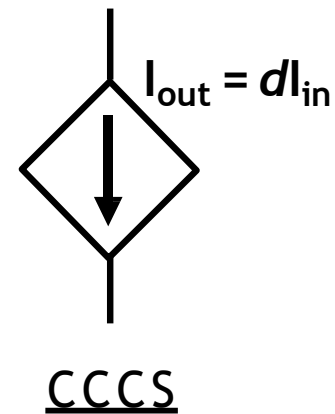
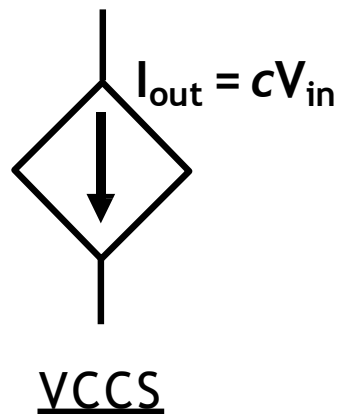
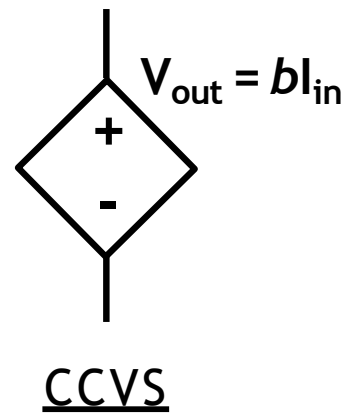
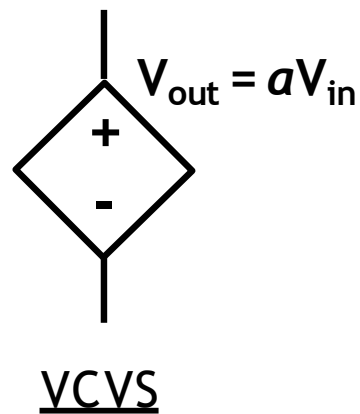
$$I_L = \frac{R_S}{R_{LOAD} + R_S} * I_s$$

$$1 = \frac{R_S}{R_{LOAD} + R_S}$$

true only if $R_S = R_{LOAD} + R_S$
so $R_{LOAD} \ll R_S$ Ideally, $R_S \rightarrow \infty$

Other dependent source forms

They may also show up on their own



3 types of gain – voltage, current, power

Voltage

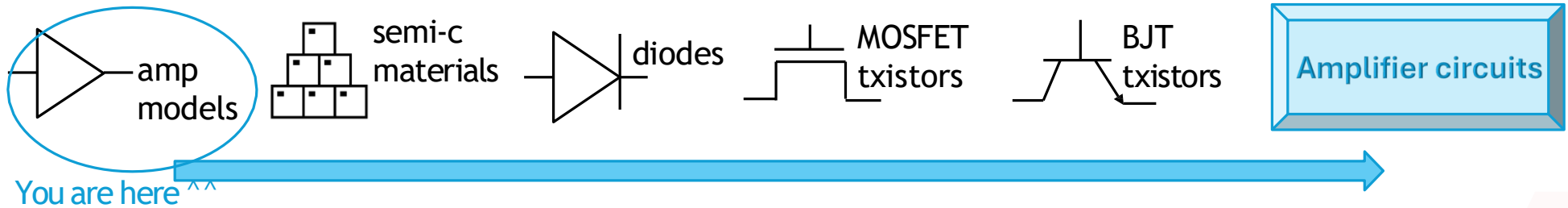
$$A_v[dB] = 20 \log |A_v| \quad [dB]$$

Current

$$A_i[dB] = 20 \log |A_i| \quad [dB]$$

Power

$$A_p[dB] = 10 \log |A_p| \quad [dB]$$



Today's Outline

End of Chapter 1, Start of Chapter 3

1. Voltage amplifier structure
2. 4 types of amplifiers
3. Intrinsic semiconductors & doping
4. Band diagrams
5. E-k diagrams



Notation conventions

Memorize this!!!

Symbol conventions

$$i_G(t) = I_G + i_g(t)$$

Diagram illustrating the notation convention for current symbols:

- $i_G(t)$ is the **Composite** symbol (indicated by a purple arrow).
- I_G is the **DC** component (indicated by a green circle and a purple arrow labeled **Uppercase**).
- $i_g(t)$ is the **AC/time-varying** component (indicated by a blue circle and an orange arrow labeled **Lowercase**).

note lower and upper cases!

Use upper case when value is:

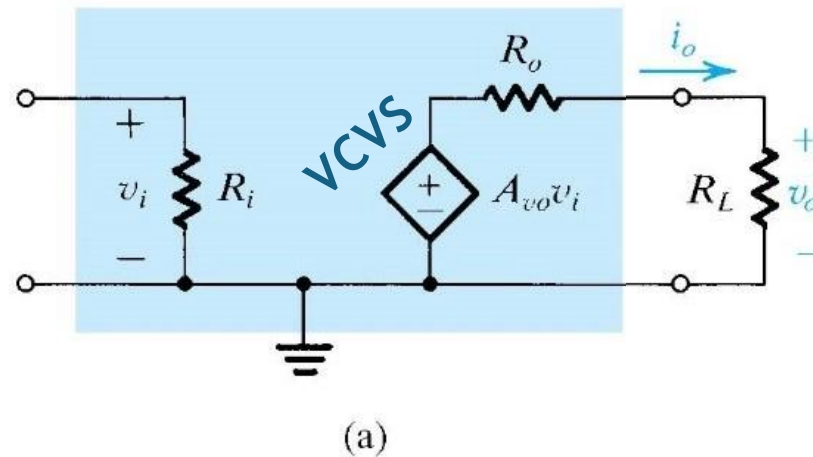
- Steady
- Mean
- RMS

Use lower case when value is:

- Instantaneous (varies with time)

e.g., $i_{\#}(t) = I_{\#}\sin(\omega t)$

Voltage amplifier construction – Sec. 15



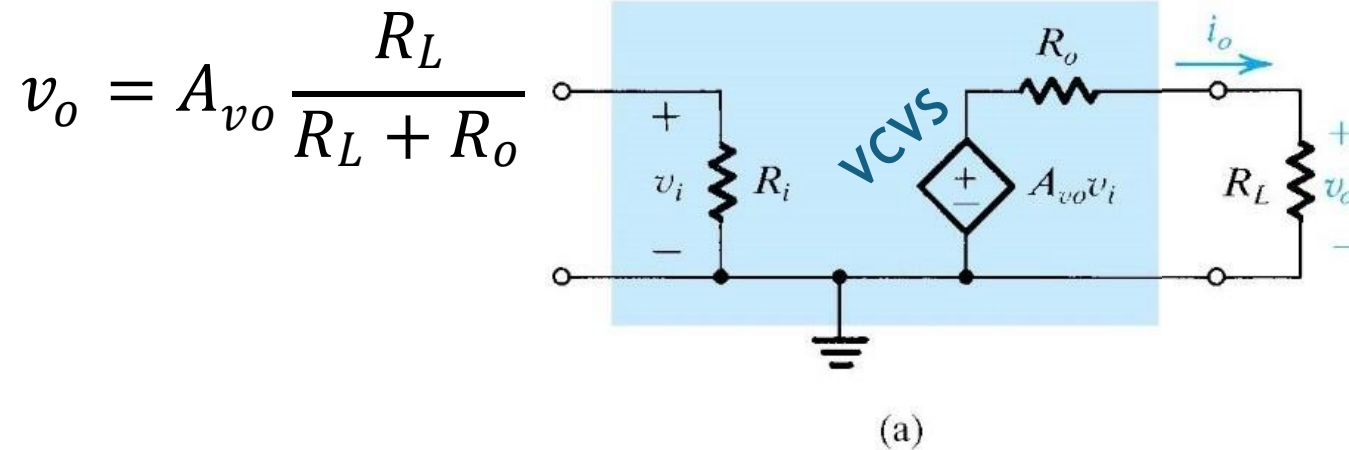
Open-circuit gain

$$A_{vo} = \left. \frac{v_o}{v_i} \right|_{R_L \rightarrow \infty}$$

or, in the general case:

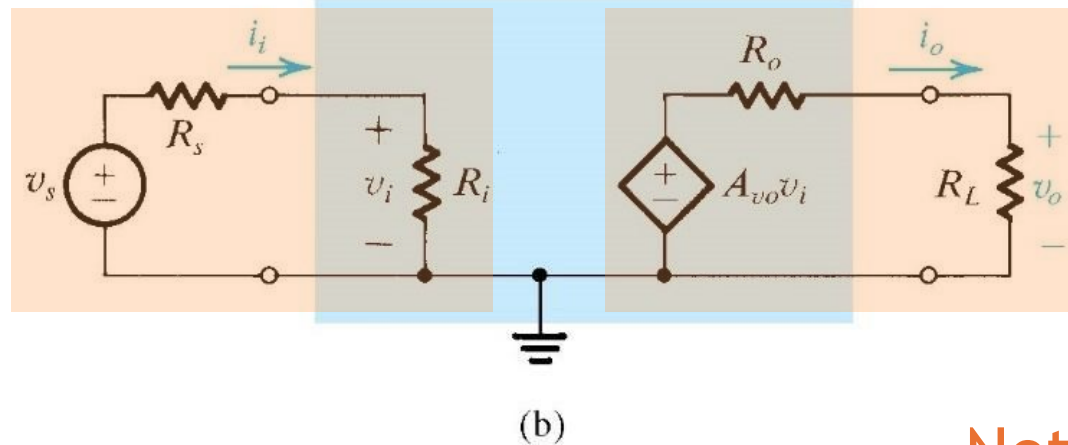
$$v_o = A_{vo} v_i \frac{R_L}{R_L + R_o}$$

Voltage amplifier construction – Sec. 15



$R_L \rightarrow \infty$ or, $R_o \rightarrow 0$
(make it as small as possible)

Voltage amplifier construction – Sec. 15



Composite voltage gain

$$\frac{v_i}{v_s} = \frac{R_i}{R_i + R_s}$$

$$\frac{v_o}{v_i} = A_{vo} \frac{R_L}{R_L + R_o}$$

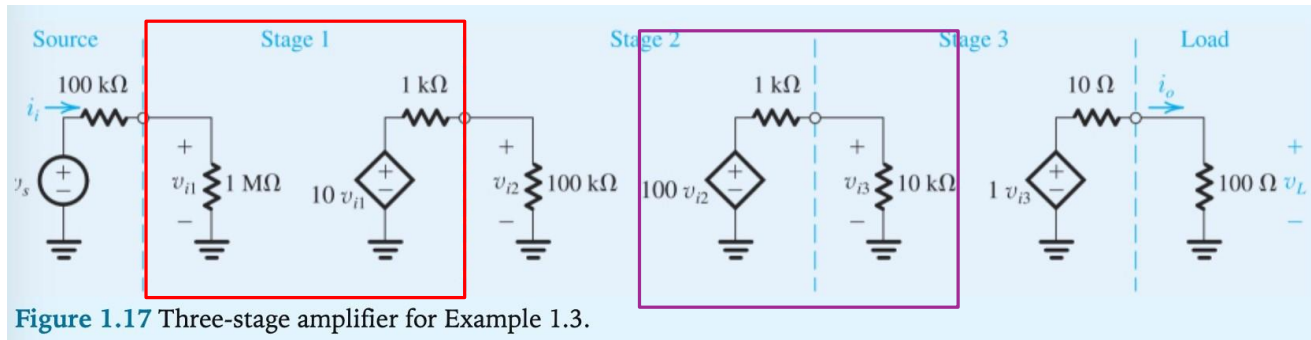
Note! $A_v < A_{vo}$

$$A_v = \frac{v_o}{v_s} = A_{vo} \frac{R_i}{R_i + R_s} \frac{R_L}{R_L + R_o}$$

Good amplifier:

$$R_i \gg R_s, \quad R_o \ll R_L$$

PSA



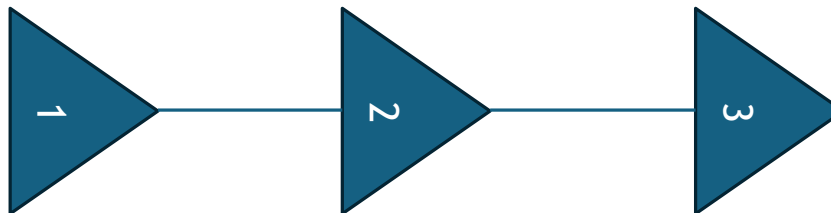
This?

1k output
resistance
of stage 1

Or this?

-> 100k load
(input resistance
of next stage)

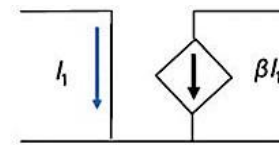
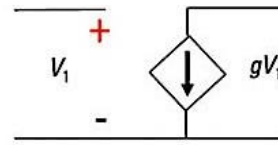
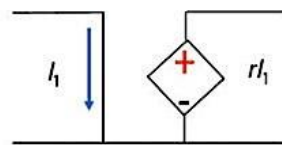
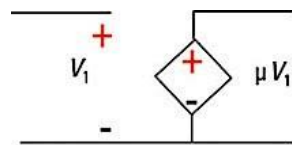
Typically, your problem will be clearly depicted in a block diagram that you can substitute the model you're used to into these components, e.g.



What is a stage?

If problems leave room for debate (e.g., as may happen on the exams), always state your assumptions! It could mean the difference between no credit and partial/full credit.

4 types of amplifiers – Table 11



Type	Circuit Model	Gain Parameter	Ideal Characteristics
Voltage Amplifier		Open-Circuit Voltage Gain $A_{vo} \equiv \frac{v_o}{v_i} \bigg _{i_o=0} \quad (\text{V/V})$	$R_i = \infty$ $R_o = 0$ VCVS
Current Amplifier		Short-Circuit Current Gain $A_{is} \equiv \frac{i_o}{i_i} \bigg _{v_o=0} \quad (\text{A/A})$	$R_i = 0$ $R_o = \infty$ CCCS

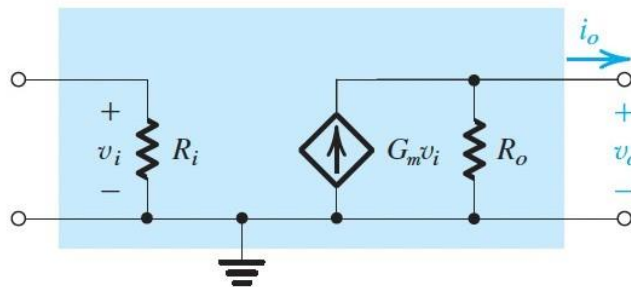
MOSFETs can be modeled as voltage amplifiers (VCVS)
 BJTs can be modeled as current amplifiers (CCCS)

These are depicted as unilateral amplifiers - signal goes from input to output (no feedback)

4 types of amplifiers (still unilateral)

Always keep in mind whether your signal is voltage or current! This will help you figure out how to connect device A to device B C make sure you meet the min specs to drive it!

Transconductance Amplifier



Short-Circuit Transconductance

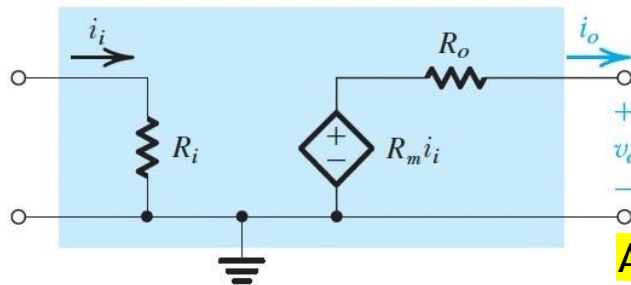
$$G_m \equiv \left. \frac{i_o}{v_i} \right|_{v_o=0} \quad (\text{A/V})$$

$$R_i = \infty$$

$$R_o = \infty$$

VCCS

Transresistance Amplifier



Open-Circuit Transresistance

$$R_m \equiv \left. \frac{v_o}{i_i} \right|_{i_o=0} \quad (\text{V/A})$$

$$R_i = 0$$

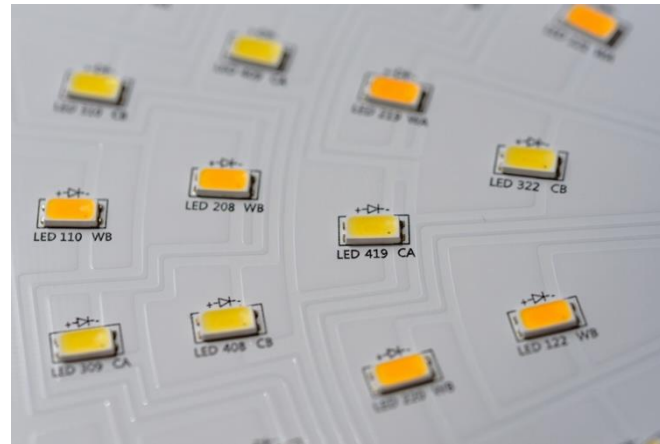
$$R_o = 0$$

CCVS

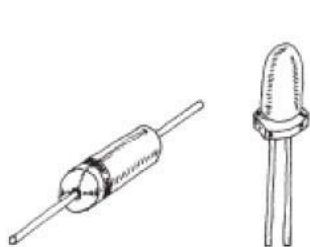
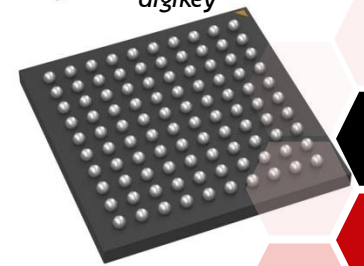
Also called "transimpedance" amplifier

and *BJT*s can also be modeled as transconductance amplifiers (VCCS)
*MOSFET*s can be configured to be transimpedance amplifiers (CCVS)

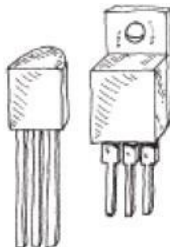
Chapter 3: Semiconductors



waveformlighting



Diodes/LEDs



Transistors



Microchips



Solar cells

(leaded components are easy to solder but not often used in consumer electronics^^)

Periodic table refresher

Group ► 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

Period ▼

Period = same number of electron orbitals

Group IV = elemental semiconductors (Si, Ge)
 Group III-V = compound (GaAs, AlGaAs, InP)
 Group II-VI = compound (ZnSe, CdTe)

Some elements near the dashed staircase are sometimes called *metalloids*

Nonmetals	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	1																	Noble gases
	H																	He
Metals	2	3	4										5	6	7	8	9	10
		Li	Be										B	C	N	O	F	Ne
	3	11	12										13	14	15	16	17	18
		Na	Mg										Al	Si	P	S	Cl	Ar
	4	19	20										31	32	33	34	35	36
		K	Ca										Ga	Ge	As	Se	Br	Kr
	5	37	38										49	50	51	52	53	54
		Rb	Sr										In	Sn	Sb	Te	I	Xe
	6	55	56										81	82	83	84	85	86
		Cs	Ba	La to Yb									Tl	Pb	Bi	Po	At	Rn
	7	87	88										113	114	115	116	117	118
		Fr	Ra	Ac to No									Nh	Fl	Mc	Lv	Ts	Og

s-block
(plus He)

f-block

d-block

p-block (excluding He)

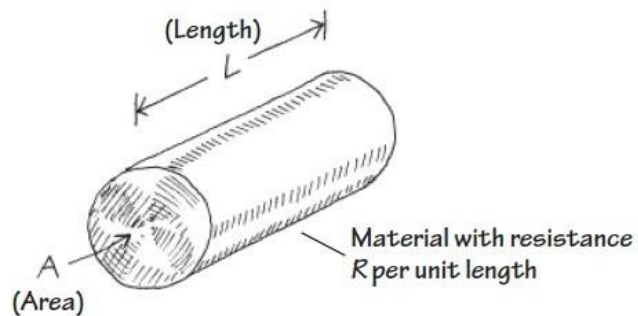
Lanthanides

Actinides

57	58	59	60	61	62	63	64	65	66	67	68	69	70
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb
89	90	91	92	93	94	95	96	97	98	99	100	101	102
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No

wiki

How do you know it's a semiconductor?

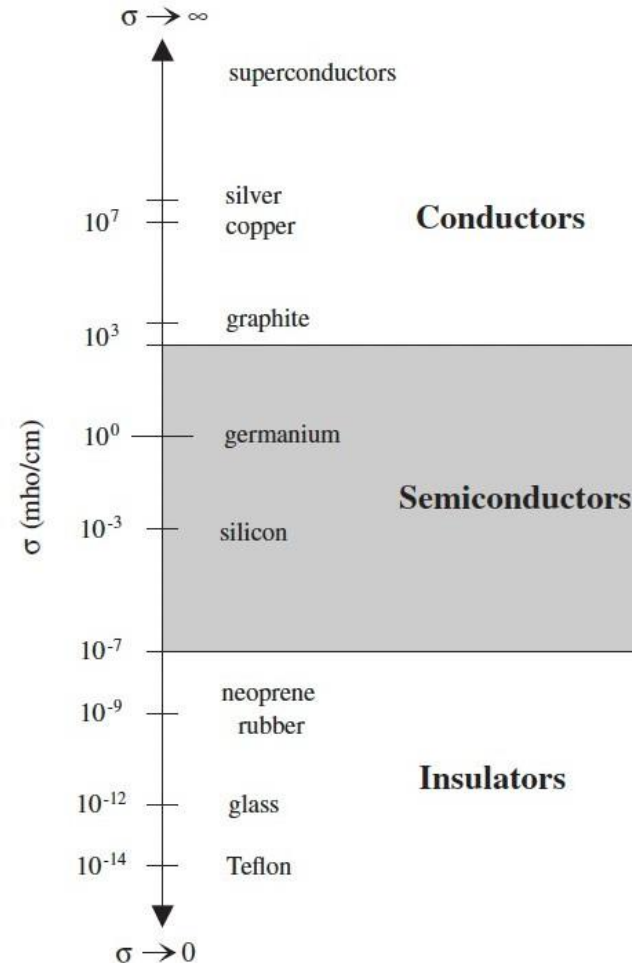


$$\rho = R \frac{A}{L} \text{ (Resistivity ohm}\cdot\text{cm)}$$

$$\sigma = \frac{1}{\rho} \text{ (Conductivity mho/cm)}$$

$$\text{mho} = \frac{1}{\text{ohm}} = \frac{1}{\Omega} = \mathcal{U}$$

FIGURE 4.2



The Bohr model - Si

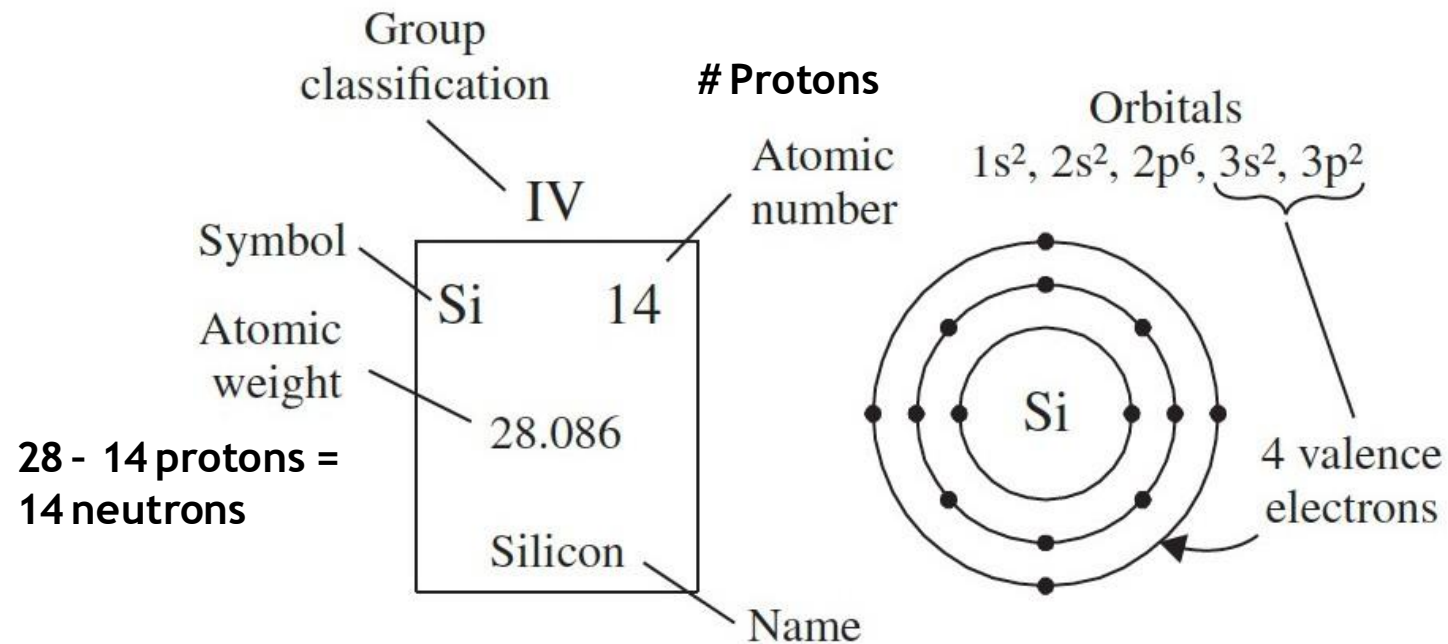


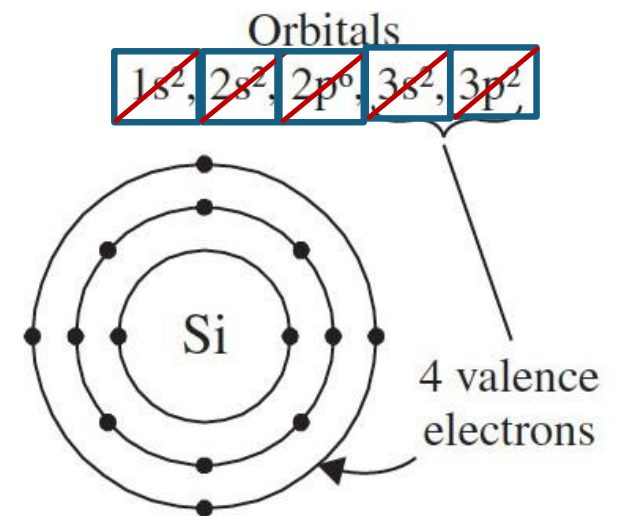
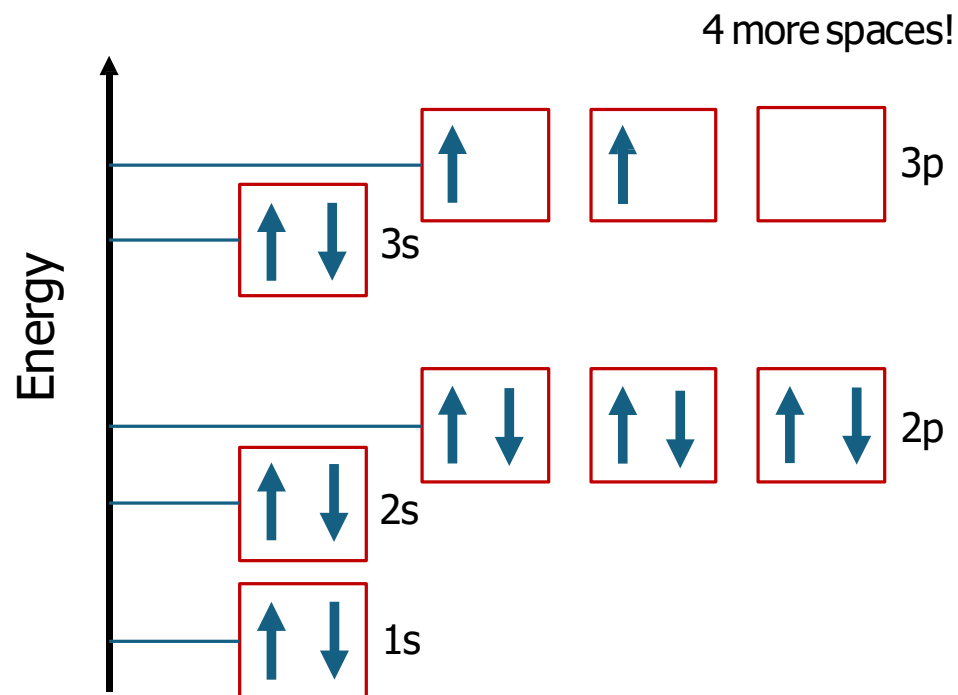
FIGURE 4.3

(Practical electronics for inventors)

Orbital energy levels

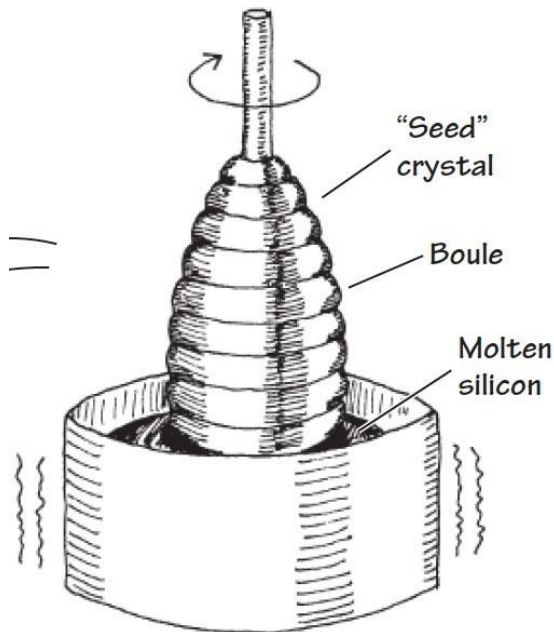
Hund's rule states that:

1. Every orbital in a sublevel is singly occupied before any orbital is doubly occupied.
2. All of the electrons in singly occupied orbitals have the same spin (to maximize total spin).



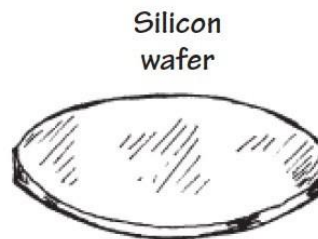
How do we process Si? (monocrystalline)

Step 1: Add SiO₂ (e.g., Quartz) to crucible



Step 3: Submerge a seed crystal and slowly pull (this allows the atoms to self-arrange/crystallize)

Step 4: Grind to smooth the diameter and slice



Heated to
~2570F ~1500C

Step 2: Heat it up, use carbon $\Rightarrow$ Si + CO₂ (byproduct)

<< What your mom thinks you do...

vs. What you really do:

https://youtu.be/5eVsQSn_EWc?feature=shared

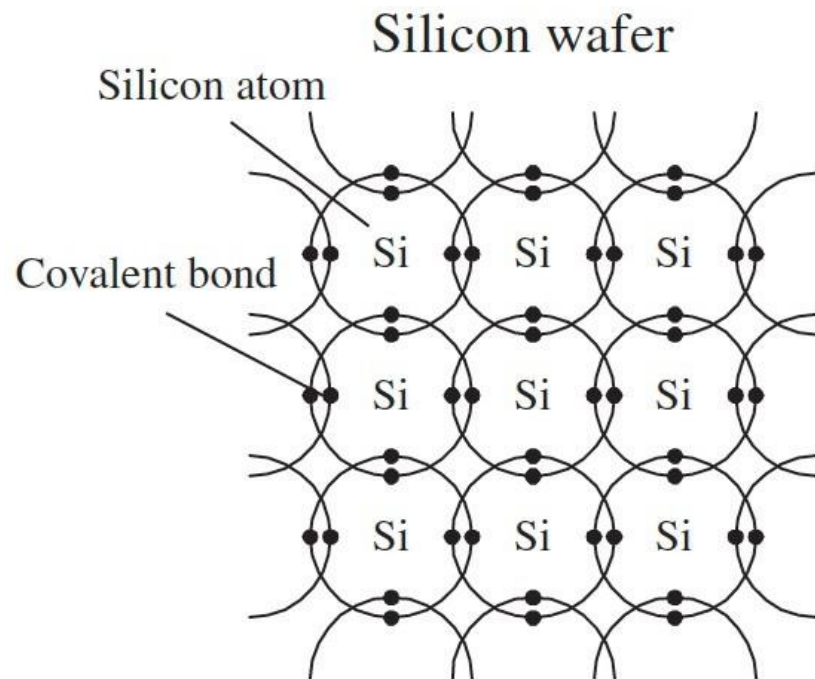
(don't try this at home)

How do we process Si?

(more to follow)



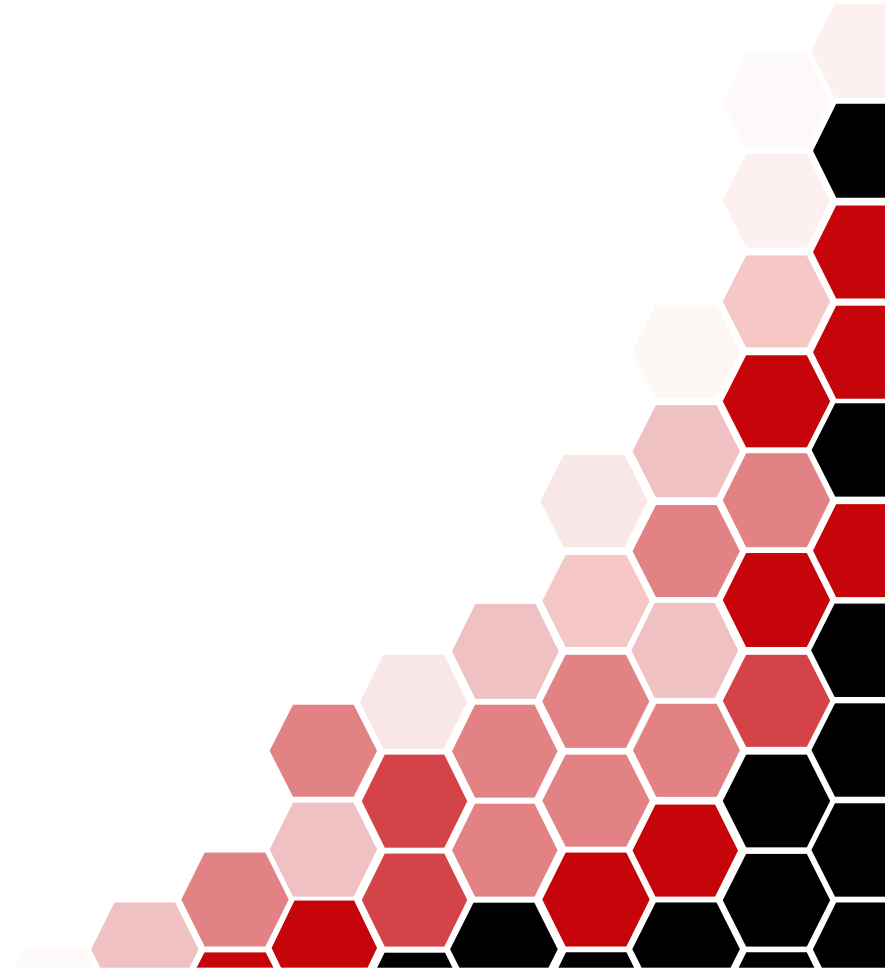
Si in bulk



Summary

- ✓ Recap of Lec. 2
- ✓ Amplifier types

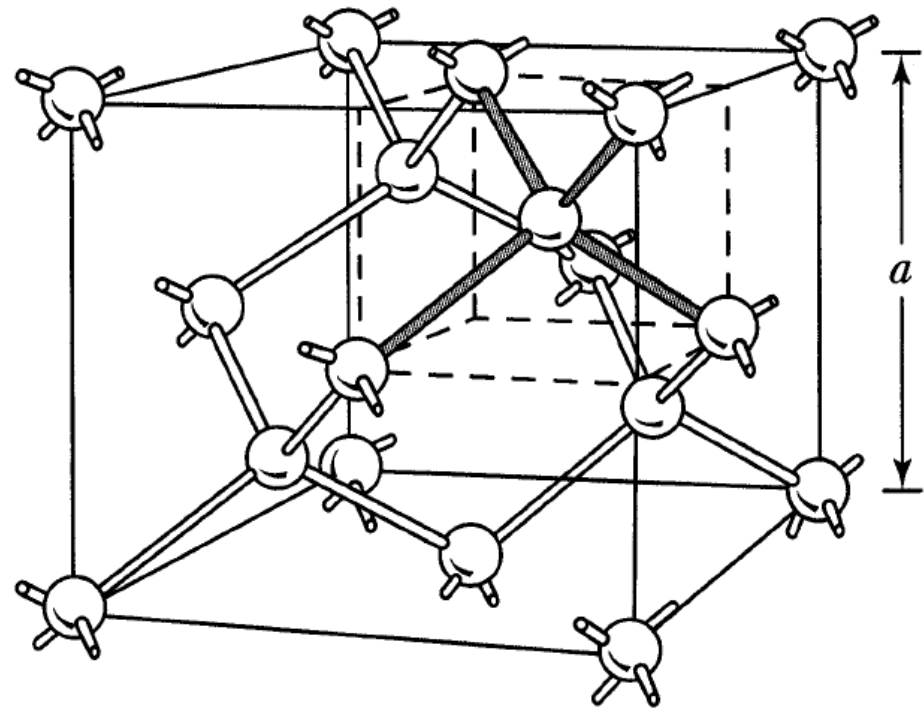
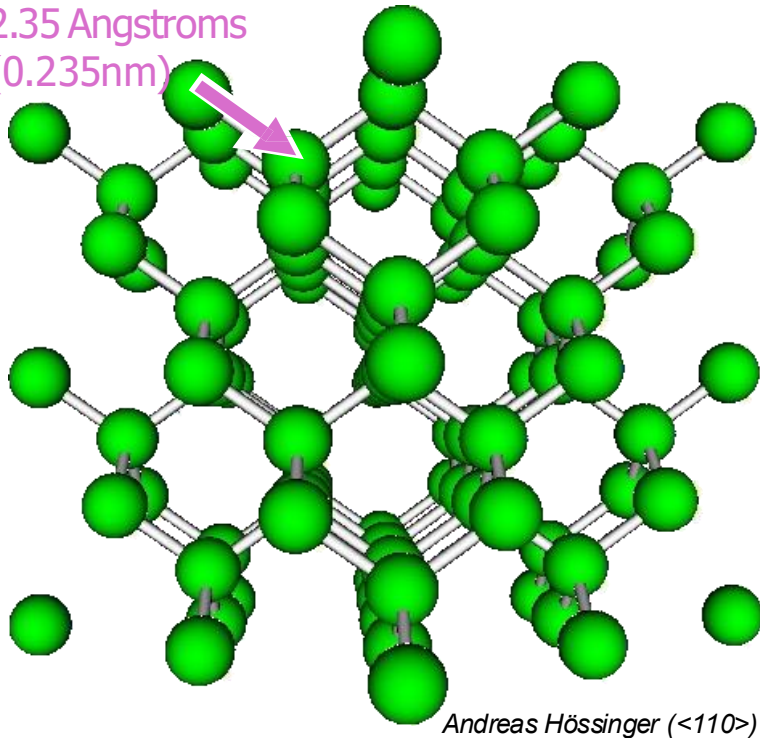
That's all, folks!



Silicon crystal structure – the “lattice”

Diamond-centered cubic

2.35 Angstroms
(0.235nm)

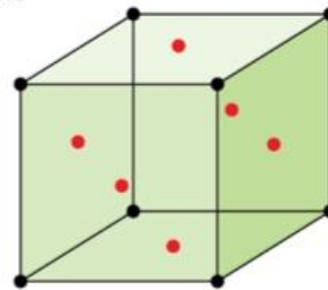
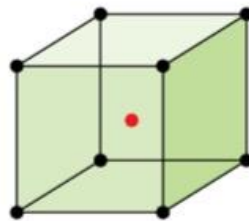
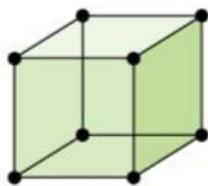


Roughly 5×10^{22} atoms/cm³

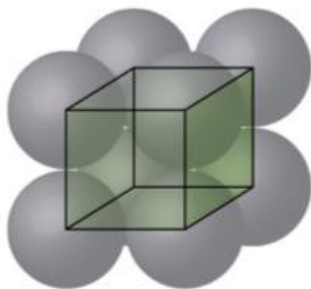
Other crystal structures

(The space here between atoms is only for diagram purposes; typical packing factors range from 34-75%)

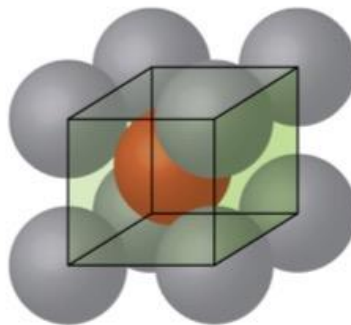
Lattice point locations



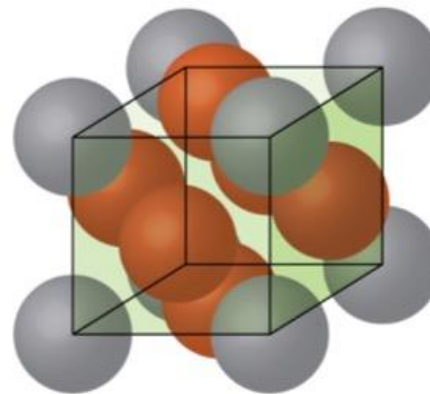
Cubic unit cells



Primitive cubic

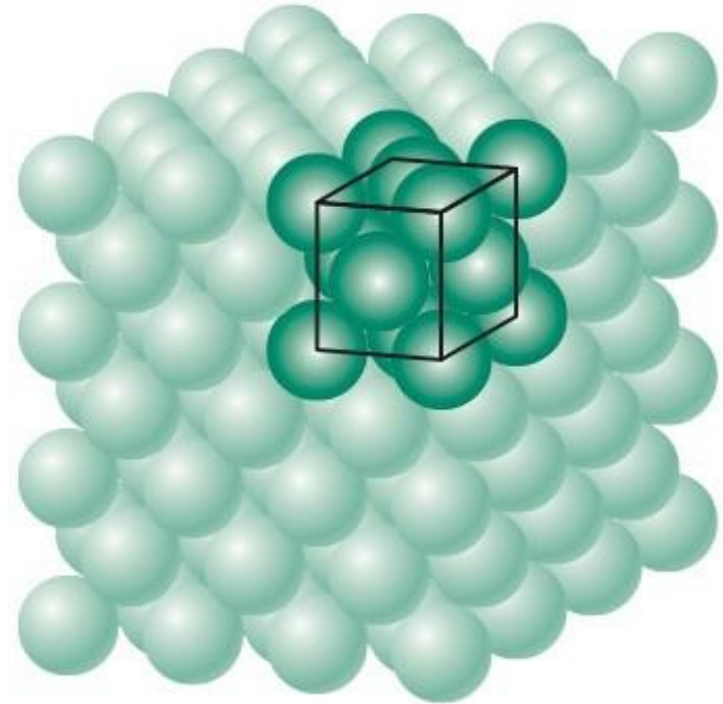
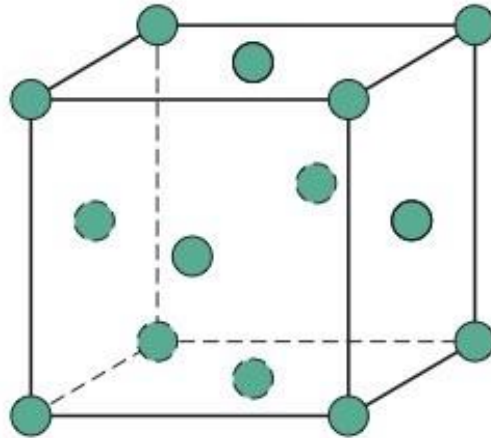
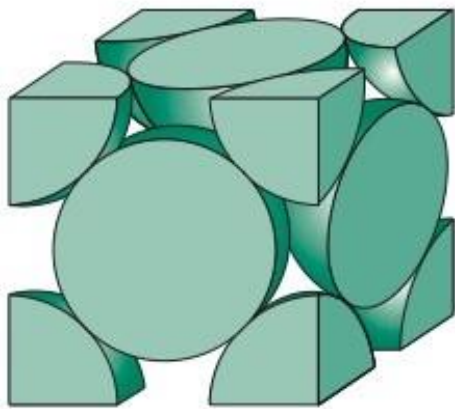


Body-centered cubic



Face-centered cubic

Example: face-centered cubic (FCC)

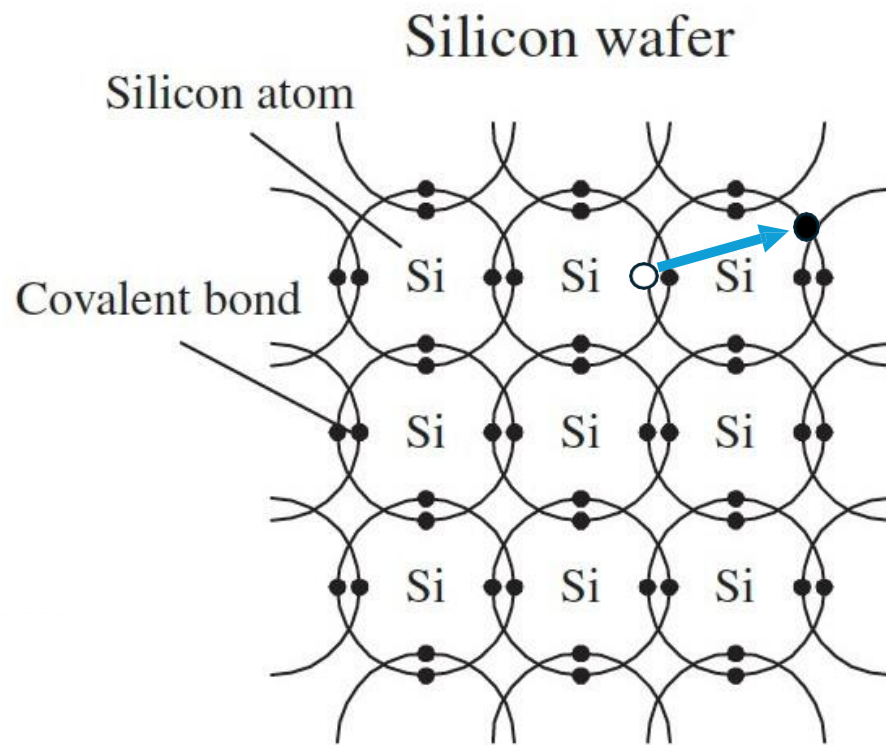


Penn State

Packing factor = 74%



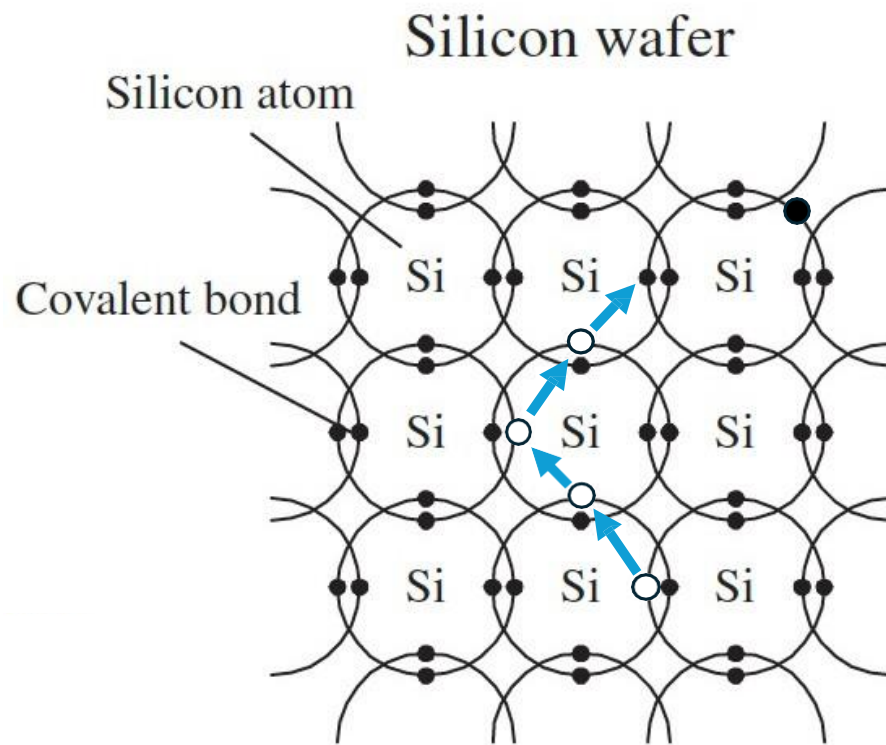
Si in bulk



A silicon wafer in pure form doesn't contain any free charge carriers; all the electrons are locked up into covalent bonds between neighboring atoms.

(At zero degrees Kelvin, they are locked up completely; at higher temps, they may jump out and jump back in)

Si in bulk



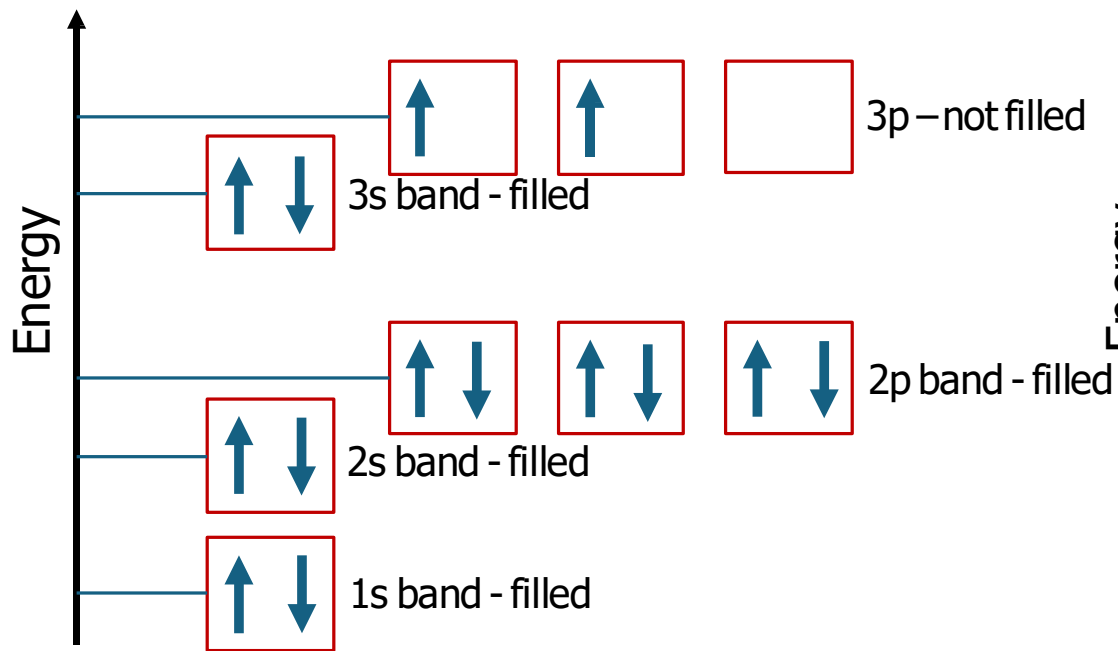
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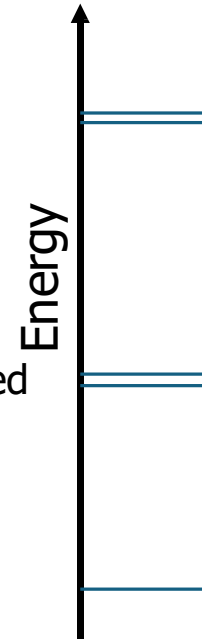
- or they may jump out, leaving what is called a "hole", which is a positive "charge carrier"
- other electrons will jump into the hole, allowing the positive charge to move around.

Energy band diagram

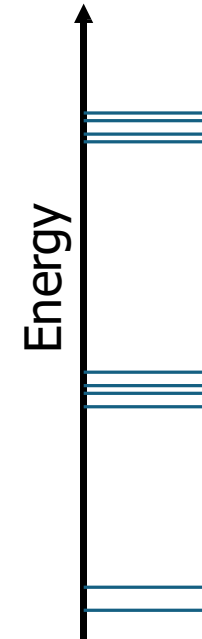
Energy is quantized for few atoms



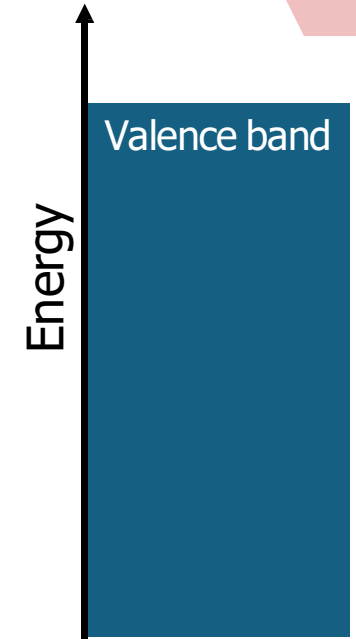
1 Si atom



1 Si atom

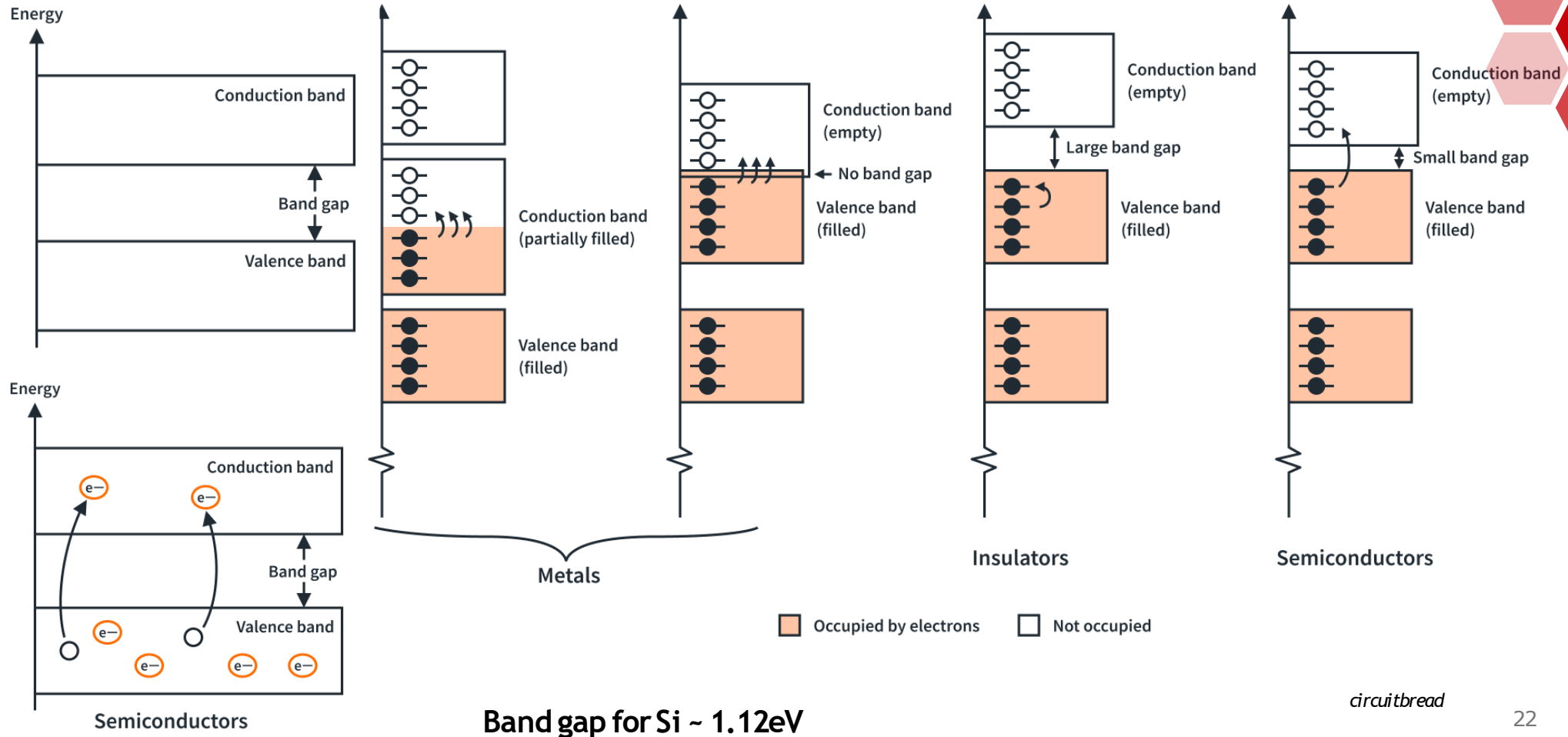


2 Si atoms

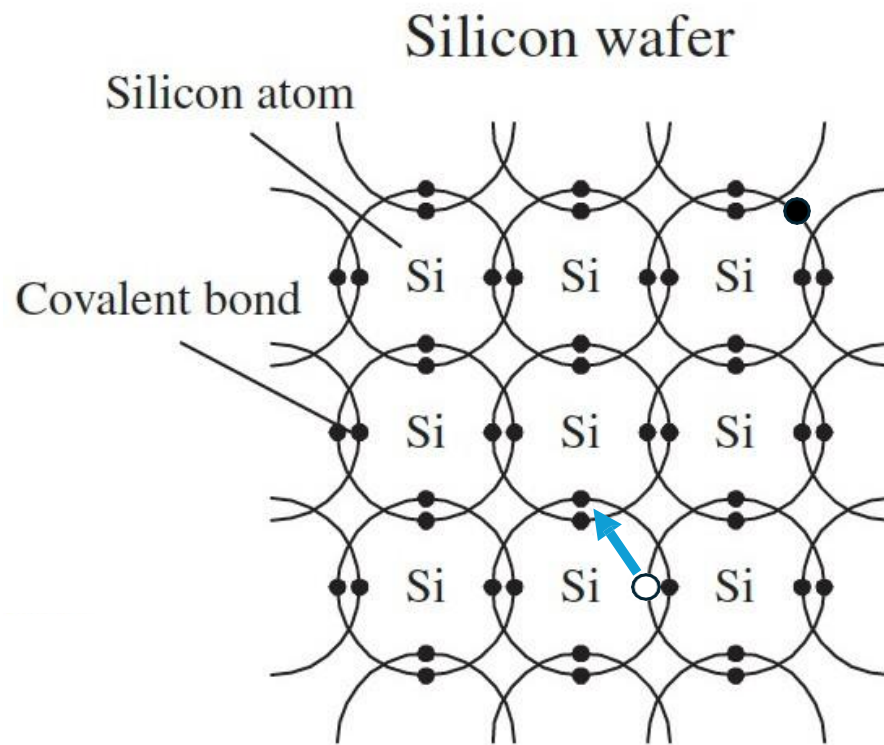


Si crystal

Band diagram



Si in bulk



When an electron moves, a hole takes its place

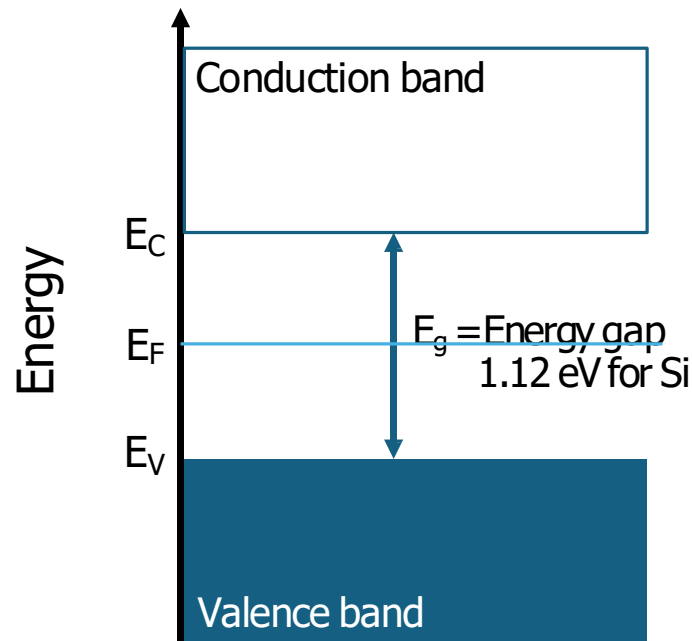
A silicon wafer in pure form doesn't contain any free charge carriers; all the electrons are locked up into covalent bonds between neighboring atoms.

(At zero degrees Kelvin, they are locked up completely; at higher temps, they may jump out and jump back in)

- or they may jump out, leaving what is called a "hole", which is a positive "charge carrier"
- other electrons will jump into the hole, allowing the positive charge to move around.

But still, net charge is zero; we don't have a useful device with just plain Si...

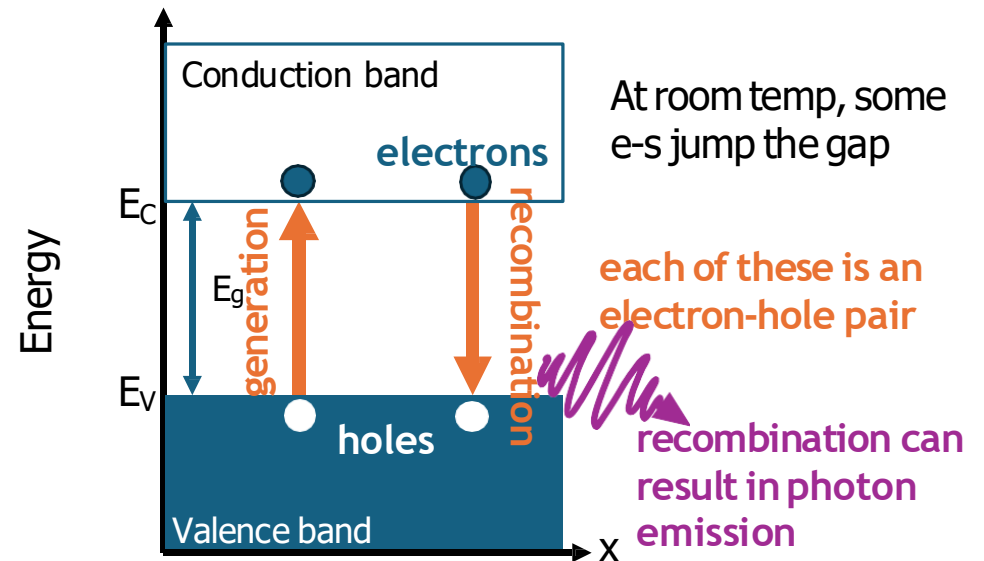
Band diagrams – more detail



intrinsic Si crystal

No electron states in the gap! But because they tunnel from valence to conduction and back, there is a nonzero probability of finding electrons there.

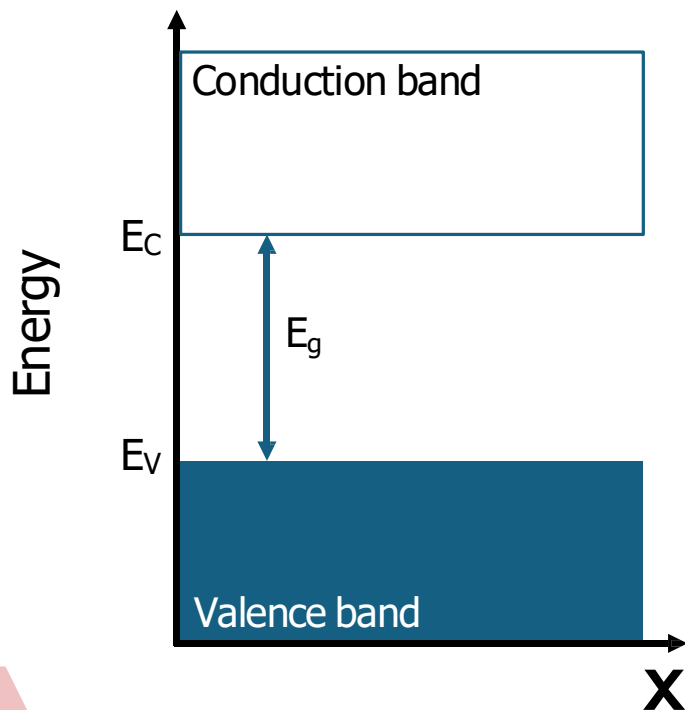
At thermodynamic equilibrium, the Fermi energy level has a 50% probability of being occupied at any given time.



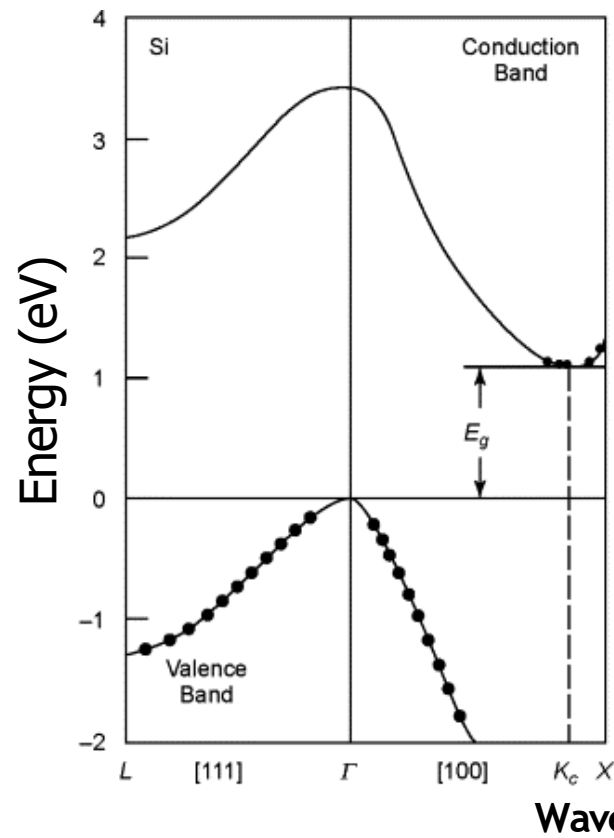
1 eV = 1.6×10^{-19} J, the energy acquired when an electron (e^-) falls thru a potential of 1 V

E-k diagrams – indirect band gap

Used for materials characterization, based on momentum of free electrons within the material. Details are beyond the scope of this course but check out the references below for more!



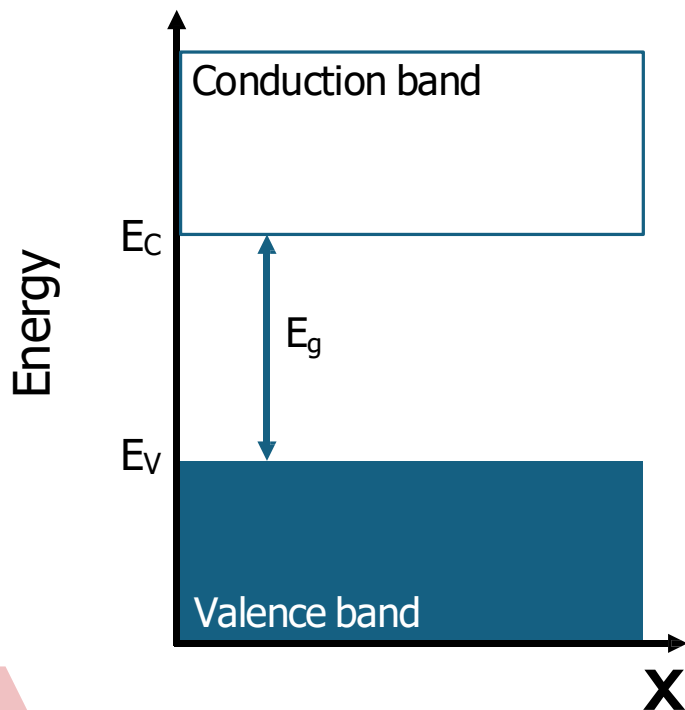
(Our band diagram)



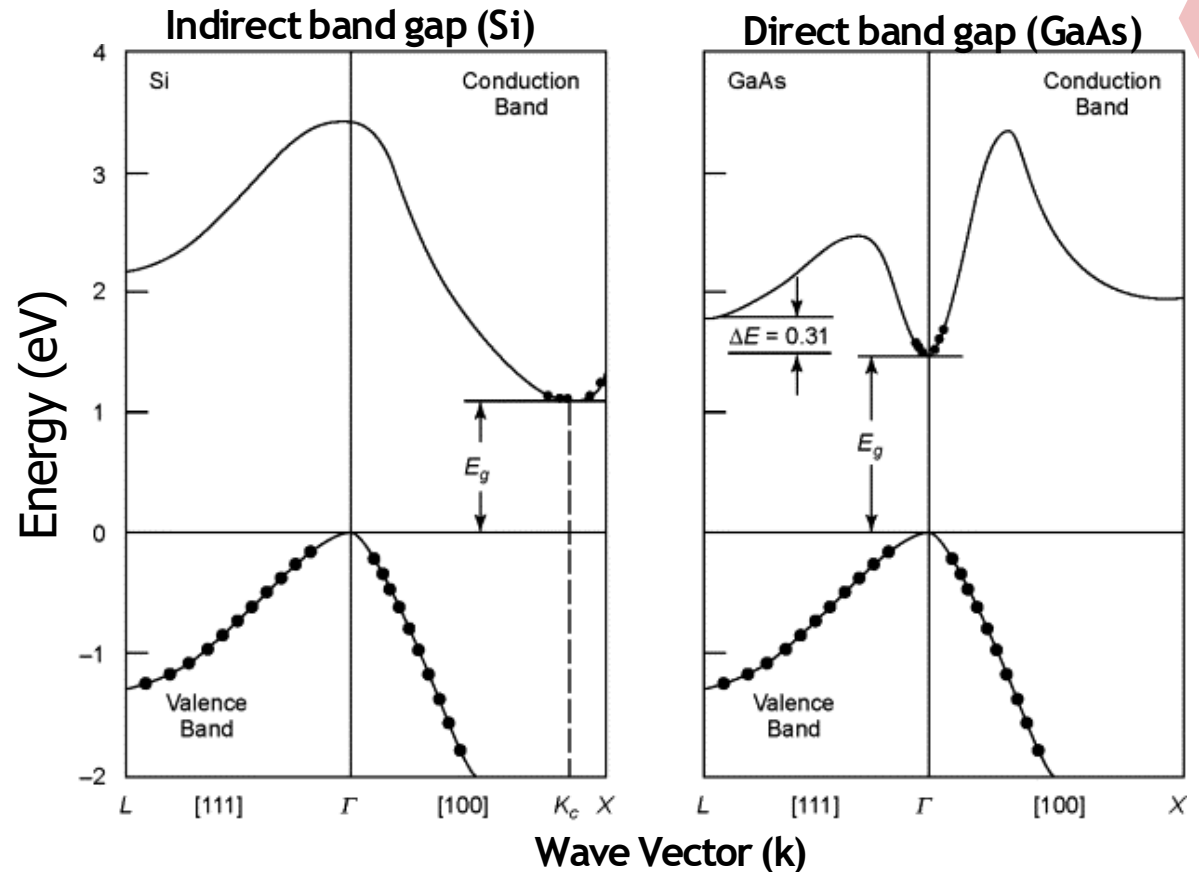
At 0 K the conduction band is empty! At room temp, there is spurious generation and recombination

E-k diagrams – direct band gap

Used for materials characterization, based on momentum of free electrons within the material. Details are beyond the scope of this course but check out the references below for more!



(Our band diagram)



<https://www.sciencedirect.com/topics/engineering/energy-band-theory>